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Explainable machine learning models for mortality prediction in patients with sepsis in tertiary care hospital ICU in low- to middle-income countries

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Abstract

Introduction Mortality in sepsis patients remains a challenging condition due to its complex nature. It is an even more prevalent health problem in low- and middle-income countries demanding costly treatment and management. This study proposes an explainable artificial intelligence-based approach towards mortality prediction for patients with sepsis admitted to intensive care unit (ICU).

Methods A total of 500 patients ($N = 500$, male: female = 262:238, age = 45.96 ± 20.92 years) with sepsis were analyzed retrospectively. We utilize SHapley Additive exPlanations (SHAP) method to gain insights into the preliminary model's learnings regarding the wide array of demographic, clinical, radiological, and laboratory features. The clinical insights were used for feature selection to fetch the top $t = 80\%$ feature spread as well as to derive empirical findings from feature dependence plots which could find application in periphery hospital settings. Four machine learning algorithms, Random Forest, XGBoost, Extra Trees and Gradient Boosting classifiers were trained for the binary classification task (discharge from ICU and death in ICU) with the selected influential feature set.

Results The Extra Trees Classifier showed the best overall performance with AUROC score: 0.87 (95% CI 0.80–0.93), Accuracy: 0.79 (95% CI 0.71–0.86), F1 score: 0.78 (95% CI 0.69–0.86), Precision: 0.88 (95% CI 0.78–0.98) and Recall: 0.70 (95% CI 0.57–0.82). All four models perform significantly well on dataset with AUROC scores ranging from 0.81 (CI 0.73–0.89) to 0.87 (CI 0.80–0.93) and F1 scores ranging 0.74 (CI 0.64–0.83) to 0.78 (CI 0.69–0.86) on the hold-out test set and were stable over fivefold cross-validation prior to testing.

Conclusions The proposed approach could provide preemptive estimations into prognostication and outcome prediction of patients with sepsis in low-resource settings. This will aid in clinical decision-making, resource allocation and research for new treatment modalities.

Keywords Critical care, Explainable artificial intelligence, Machine learning, Mortality prediction, Sepsis, ICU

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Introduction

Sepsis has a non-uniform nature and is a complex syndrome of physiologic and biochemical abnormalities. With its complex characteristics, involving organ dysfunction, it can lead to severely increased chance of patient mortality in cases of improper management of sepsis patients in hospital settings [1]. Even reliable measuring of sepsis incidence and trends is a challenging task [2]. It is estimated that in 2017, about 48·9 million sepsis incidences were recorded worldwide and around 11·0 million sepsis-related mortalities (around 19·7% of all global deaths) [3]. Due to several factors such as growth in clinical awareness, changes made in diagnostics and coding protocols and non-standardized definitions, there is an uncertainty as to the accuracy of reported trends as well as marked heterogeneity in incidence and mortality rates [2, 4–8]. This is also a significant problem in terms of the cost of treatment and management. Moreover, the mortality due to sepsis has a higher occurrence in low- and middle-income countries/low-income counties (LMICs/LICs) with respect to high-income countries (HICs) [9]. The highest burden of sepsis was noted in the Global Burden of Disease study to be in LMICs accounting for 85% of total sepsis-related deaths worldwide [10]. Therefore, preemptive estimation into the septic patient's outcome could greatly assist clinical decision-making to prepare in terms of treatment planning, resource allocation as well as cost estimation.

Sequential Organ Failure Assessment (SOFA) [11] and Acute Physiology and Chronic Health Evaluation II (APACHE-II) [12] are the two most common scores used to stratify sepsis patients. However, these severity scoring systems were created and validated long ago and the validity of this score primarily stems from analyzing over a million ICU electronic health records of critically ill patients with suspected sepsis in HICs. The differences in patients, pathogens, and clinical management capacity between HICs and LMICs are substantial. Consequently, it remains uncertain whether these scores are applicable across various types of infections, hospital locations, and countries [13]. Although, the conventional severity score showed fair discriminatory ability for predicting ICU mortality with a range of AUROC: 0.67–0.8; artificial intelligence (AI)-based models to predict mortality in sepsis patients showed higher discriminating abilities than the severity scores showing AUROC: 0.86–0.97 [14–16]. In a comprehensive meta-analysis over 50 studies developing a total of 104 machine learning model for predicting sepsis-related death, Zhang et al. [17] demonstrated overall C-index of 0.799 (95% CI 0.779–0.819) and 0.774 (95% CI 0.763–0.785) in the training and validation sets, respectively, for ML models, while overall C-index of the severity scoring systems was 0.717 (95% CI 0.673–0.761) and 0.689 (95% CI 0.633–0.745), respectively.

Therefore, there is an inherent need to create appropriate accountable data-driven approaches in emergency healthcare. The goal of such studies is to eventually serve as AI-based clinical decision support systems with the clinical community being the interactive user. However, adoption and credibility of such tools get severely hampered by reasons such as its black-box nature and lack of domain knowledge as well as interpretability [18]. Explainability plays a significant role in terms of technological, legal, medical, and patient perspectives [19]. Moreover, this may further serve towards alignment of model performance with clinical protocols and objectives [20].

While there have been several developments with regard to studies predicting in-hospital sepsis-related mortality utilizing machine learning (ML) and deep learning (DL) algorithms [15–32], most of such contemporary studies [21–24, 26, 28–31] are based around the large Multiparameter Intelligent Monitoring in Intensive Care (MIMIC) database [33]. While being a comprehensive dataset, the data in the MIMIC database are heterogeneous and high dimensional with over 18,000 collected variables [34]. Moreover, the outcome variable: in-hospital-mortality is imbalanced (~ 1:9 ratio of deaths to survivors) which may hamper the performance of predictive models [35]. Fewer studies involved in-hospital ICU data [15, 27] and emergency department data [16, 25, 32] from HICs. Throughout the literature, to our knowledge, we observed a lack of studies specifically catering or extending application to LMIC environments.

Feature selection is another very critical area which becomes essential due to high dimensional healthcare datasets and associated variability in the features [36]. Feature selection essentially reduces dimensionality of the input data and can improve model performance by multiple ways such as decreasing the learning speed, reducing model complexity or by increasing generalization capacity and classification accuracy [37]. Selection of suitable representative features can significantly reduce computational costs and lead to greater understanding of the problem. Another observed shortcoming was, there seem to be vary less studies [15, 22] dedicated to interpretability and explainability of models used for feature selection as well as integration of clinical insights.

With regard to the aforementioned, this research investigates a novel machine learning model-based prediction of mortality in patients with sepsis in ICU settings. A twofold utilization of explainable AI study was employed: firstly, feature engineering and selection and secondly, for generating data-driven insights into the model's learnings using SHapley Additive exPlanations (SHAP) [38].

Finally using the selected features, a comparative set of ML models were trained to predict mortality through a binary classification task.

Materials and methods

Dataset

The study was approved by the Institute Ethic Committee of the All India Institute of Medical Sciences (AIIMS), New Delhi, India, with ethical approval number, IECPG-190/20.04.2023, RT-05/07.06.2023. A retrospective dataset of septicemia patients, admitted to different Intensive Care Units (ICUs) in AIIMS New Delhi, India, from January 2021 to January 2024, was collected from ICU clinical records. The inclusion criteria were patients with more than 18 years of age and diagnosed with sepsis (according to Sepsis-3 definition [39]) during admission to ICU. The exclusion criteria were patients with non-septic shock and already developed refractory septic shock during ICU admission. The outcome of the patient, i.e., mortality or discharge (discharge from ICU) was taken as the target variable (in binary variable format). A total of 500 patients (N= 500, male: female =262:238, age =45.964 ±20.921 years) were randomly included in the study taking 250 patients each in the mortality and discharge group to match the mortality–discharge ratio 1:1. Clinical data of patients were collected only once within the first 24 h of admission to the ICU and directly entered into an Excel sheet along with the demographic data for further analysis. The dataset contained total 108 parameters comprising patient demographics, clinical, ventilatory, blood-gas, laboratory, respiratory and cardiovascular parameters; information about source of sepsis, organ dysfunction, metabolic disorders; interventions, regarding oxygenation, vasopressors, antibiotics and some miscellaneous parameters like duration of illness to ICU stay, etc. Details of different parameters are presented in Supplementary Table 1. Patient demography and salient clinical parameters in mortality and discharge groups are presented in Table 1.

Preprocessing

Data were first cleaned by removing duplicate and irrelevant observations, and structural errors to ensure uniformity of data across each clinical parameter. Textual information in terms of antibiotics and procedures invoked were converted into categorical and count-based variables by clinical inputs. The antibiotic groups were clinically engineered and categorized. The patient data acquired were clinically evaluated to remove any redundant features. The nature of data was mixed including continuous, categorical, and binary data points. One-hot encoding of categorical data points, i.e., conversion of categorical data from string or numeric values to binary values across each category was performed. To handle the missing values, features with more than 30% missing values were excluded from the dataset. For dichotomous features, missing data were replaced with mode of respective feature across the sample and for continuous valued features, Multivariate Imputer with Bayesian Ridge Estimator (https://scikit-learn.org) was used to replace the missing value in the respective feature across the sample. The features with missing values that were considered for preprocessing are listed in Supplementary Table 2. After preprocessing, a total of 138 features were used for further analysis.

Correlation and predictive power score analysis

Pearson correlation analysis was performed on the resulting features as displayed in Supplementary Fig. 1. Similarly, predictive power scores were determined to investigate the nonlinear feature relations. It is to be noted that most features did not show any significant correlation or predictive power scores. Therefore, all 138 features were retained for further analysis [36, 37].

Feature analysis using explainable AI

Machine learning-based models are usually considered a black-box by the clinical community. They may present efficient solutions, but do not prove much justification as to the implicit learnings from the data and are usually selected based on their performance rather than interpretability and practical insights [40]. Since we are dealing with a high dimensional dataset, SHAP [38] was utilized to provide local and global insights with regard to the model output providing the most influencing features. SHapley Additive exPlanations (SHAP) is a game theory based model interpretability approach that provides post hoc estimations of the conditional impact of a feature on the model's prediction with respect to all other features [41]. SHAP methodology is model agnostic and can be used across different machine learning models for comparison. It essentially computes the sum of feature contributions for each instance to give an explanation of the predictive output. It determines a value (SHAP value) that signifies the role of each feature in the model's optimal outcome and ranks the features in order of importance. These values are useful for assessing the significance of individual features and its impact on the model's outcomes. Specifically, here the Tree Explainer [42] was used with the baseline estimator as XGBoost [43] that utilizes previous models' residuals to create new models iteratively to make predictions. It has been noted to outperform even deep learning models for tabular data and does not require any significant fine-tuning for the dataset [44].

Table 1 Patient characteristics and baseline clinical and laboratory parameters

Parameters	Mortality group ($n = 250$)	Discharge group ($n = 250$)	*p*-value
Demographic parameters			
Age	46.5 [7–93]	40.5 [16–100]	0.334
Sex—female	102 (40.8)	136 (54.4)	0.003
Weight	65 [20–110]	65 [33–170]	0.533
No substance abuse	161 (64.4)	192 (76.8)	0.003
Smoking	73 (29.2)	41 (16.4)	0.001
Alcoholism	63 (25.2)	40 (16)	0.015
Other substance abuse	26 (10.4)	11 (4.4)	0.017
Duration of illness before ICU admission	13 [1–365]	7 [1–120]	$< 10^{-3}$
Clinical parameters			
Glasgow Coma Scale (GCS)	9 [3–15]	11 [3–15]	$< 10^{-3}$
Heart rate	117 [45–183]	100 [48–180]	$< 10^{-3}$
Blood pressure systolic	110 [60–190]	121 [11–200]	$< 10^{-3}$
Blood pressure diastolic	64 [30–120]	72 [40–110]	$< 10^{-3}$
Respiratory rate	25 [12–40]	24 [12–732]	0.851
Saturation of peripheral oxygen (SpO2)	98 [66–1010]	99 [84–100]	0.527
Intubated	198 (79.2)	129 (51.6)	$< 10^{-3}$
Lactate (mmol/L)	2.5 [0.11–19.3]	1.2 [0.24–9.4]	$< 10^{-3}$
pH	7.32 [6.7–8]	7.39 [6.92–7.57]	$< 10^{-3}$
PaO2/FiO2-ratio	193 [44–492]	232.15 [44.8–495.8]	0.026
Hb (g/dl)	8.8 [3.6–12.12]	9.6 [4.5–11.5]	0.463
TLC (/mm^3)	12,920 [0.71–116600]	12,140 [9.62–85550]	0.993
PLT (/mm^3)	1.03 [0.07–70]	1.6 [0.07–77]	0.669
PT (s)	17.75 [10.2–41.7]	14.2 [8.8–34.9]	$< 10^{-3}$
INR	1.54 [0.8–3.8]	1.2 [0.45–2.6]	0.001
APTT (s)	41.61 [17.8–88]	33.4 [1.52–75]	0.008
Urea (mg/dl)	72.5 [2–318]	48.5 [7–326]	$< 10^{-3}$
Creatinine (mg/dl)	1.8 [0.3–12.2]	1 [0.2–13]	0.064
Albumin (g/dl)	2.45 [1–4.7]	2.8 [1.4–5.6]	$< 10^{-3}$
SOFA score	10 [1–20]	6 [2–16]	$< 10^{-3}$
Source of sepsis			
Pulmonary	169 (67.60)	159 (63.6)	0.397
Intra-abdominal	48 (19.2)	36 (14.4)	0.188
Genito-urinary	25 (10)	36 (14.4)	0.172
Surgical site infection (SSI)	31 (12.4)	21 (8.4)	0.187
Central nervous system (CNS)	9 (3.6)	9 (3.6)	1.00
Other	42 (16.8)	55 (22)	0.175
Organ dysfunction			
Respiratory (PaO2/FiO2 ratio < 300)	189 (75.6)	166 (66.4)	0.03
Liver (INR > 1.5 and total bilirubin > 2 mg/dl)	136 (54.4)	73 (29.2)	$< 10^{-3}$
Hemodynamic (MAP < 65 mmHg or vasopressor requirement)	186 (74.4)	96 (38.4)	$< 10^{-3}$
Central nervous system (GCS < 15)	138 (55.2)	67 (26.8)	$< 10^{-3}$
Renal (creatinine > 1.5 mg/dl)	155 (62)	98 (39.2)	$< 10^{-3}$
Septic shock	186 (74.4)	91 (36.4)	$< 10^{-3}$

Values are presented as median [range] or total count (%). The p values were calculated using Chi-squared test for categorical parameters and t test for continuous parameters

PaO2 partial pressure of oxygen, *FiO2* fraction of inspired oxygen, *HB* hemoglobin, *TLC* total leucocyte count, *PLT* platelet count, *PT* prothrombin time, *INR* international normalized ratio, *APTT* activated partial thromboplastin time, *MAP* mean arterial pressure, *SOFA* score Sequential Organ Failure Assessment score

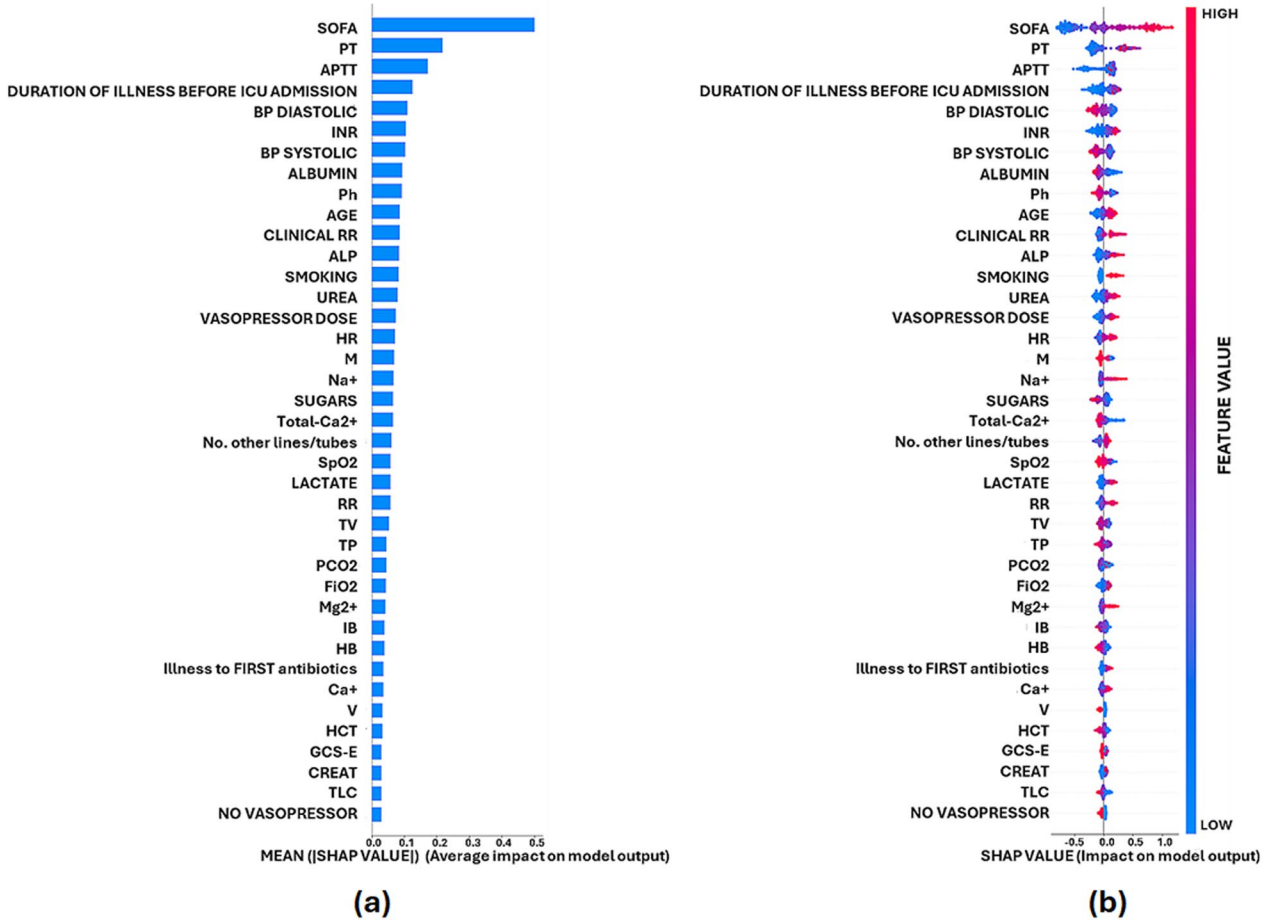


Fig. 1 Feature-wise impact of features on model output. **a** Bar chart representing mean absolute SHAP values of features. **b** Density scatter plot showing SHAP value with directional information on model output. Values towards the + ve x-axis (right) signifies the tendency of the parameter to drive the model output towards giving a mortality prediction and -ve axis towards an alive state prediction. Red signifies a higher value of the feature and blue a lower value

The complete dataset was split into training and hold-out test set in the ratio of 4:1. An XGBoost instance was trained on the former data subset considering as the baseline estimator for Tree SHAP explainer allowing for the exact computation of SHAP values for tree ensemble methods. Following these the local and global explanations were determined via SHAP. The purpose of this was twofold: first was the identification of most influential features to the model's predictive outputs as well as looking into local and global inter-feature dependencies learnt by the model. Second was identifying inter-feature relations with respect to the model. SHAP interaction plots were generated for the selected important features to visualize the impact of the individual feature towards the mortality due to septic shock among patients.

Feature selection

The mean absolute SHAP-based feature influence values were determined and plotted. The summation of these was considered the total reference value, and based on the observed spread of values, features contributing to the top 't'% (t= 80% in this study) of information were selected. These features were then considered for further model training and performance evaluation.

Model development and evaluation

With the selected features from the previous step, four tree-based models namely Random Forest Classifier, XGBoost Classifier, Extra Trees Classifier and Gradient Boosting Classifier were considered for prediction model development. Prediction models using those four algorithms were also developed with the current standard of practice SOFA scores separately for comparison with the proposed model. The training dataset comprising 400 samples was used to train the four classifiers with five-fold cross-validation. The hold-out test set of 100 patients was retained separately to ensure no data leakage and to evaluate model efficacy. Performance metrics such as

area under the receiver operating characteristics curve (AUROC) values, accuracy, F1 score, precision and recall were evaluated for each cross-validation fold to assess the training of the models. Mean values of all metrics across the fivefold cross-validation were evaluated for the four classifiers and compared. Finally, the performances of trained classifiers were evaluated on the hold-out test set and compared.

Results

Feature selection

After preprocessing, the distributions of mean |SHAP values| of the resulting influential features are shown in Supplementary Fig. 2. Clinical, laboratory and blood-gas features occupied 71% (36%, 20%, and 15%, respectively) of the total SHAP value spread. Ventilatory, intervention, demographic, dysfunction and disorder, and time duration features held the remaining 29% of the total SHAP value spread. Based on mean absolute SHAP value a subset of $n = 39$ most influential features representing 80% of total SHAP value spread were selected for further model development (Supplementary Fig. 3).

SHAP-based feature analysis

Figure 1.a shows the average SHAP value magnitudes of the selected 39 most influential features in a bar chart in descending order. SOFA score, Prothrombin time (PT), Activated Partial Thromboplastin Time (APTT) were observed to be the features having a significant impact on the model output. These were followed by the duration of illness of the patient before ICU admission, International normalized ratio (INR) as well as the systolic and diastolic blood pressure (BP).

A density scattered summary plot of SHAP values is shown in Fig. 1b, wherein, each point depicts a single sample, i.e., a single patient. The actual SHAP value (the logarithmic odds for predicting patient mortality) is represented along the X-axis where a higher SHAP value indicates a higher chance of mortality in comparison to a lower SHAP value. Similar to Fig. 1a, along the Y-axis, features are arranged based on their importance, which is given by the means of their absolute Shapley values. The color scheme represents the features' value. It can be noted that SOFA score affects a large number of predictions by a significant amount. PT follows closely after it with a similar nature. However, APTT, PT, duration of illness, INR and BP seem to affect most predictions but by a small amount.

Figure 2 depicts individual dependence plots for the top nine features from SHAP analysis and patients' demographic features, showing the effect of a single feature on the overall models' prediction performance. Some notable observations can be made from the dependency plots regarding the nonlinear effects of the features on the model output. For the SOFA score, a turning point was observed where patients with scores lower than 10 seem to be at a lower mortality risk, however, past this score the risk was found to increase steeply (Fig. 2a). Increased INR of patients was found to show a sharp push towards mortality (Fig. 2f) while, lower systolic and diastolic BP (< 120 mmHg and < 70 mmHg, respectively) show an increased risk of mortality (Fig. 2e and g). Albumin levels below ~ 2.5 mg/dl and pH under ~ 7.35 also showed a sharply increased risk for mortality among ICU patients (Fig. 2h and i). Concerning patient demographics, patients with more than 50 years of age and above 60 kg of weight were observed to be at a higher risk of mortality and the risk was found to increase with increasing age and weight (Fig. 2j and l, respectively). In our dataset, we also observed that females being at a lower risk than males (Fig. 2k) and non-smoking patients with no substance abuse and no alcoholism were at a lower risk than those partaking in such activities (Fig. 2m–o).

Predictive model performance

The performance of the prediction models using selected feature set and SOFA score alone is summarized in Table 2. Using SOFA scores alone, all four models demonstrated average cross-validation AUROC in the range 0.786 ± 0.041 to 0.792 ± 0.043 and AUROC of 0.77 (95% CI 0.67–0.86) on the hold-out test dataset. While all the proposed ML models using the selected feature set produced superior accuracies on our dataset showing average cross-validation AUROC ranging from 0.822 ± 0.015 to 0.843 ± 0.018 and F1 scores ranging 0.726 ± 0.025 to 0.779 ± 0.042 for training data and on hold-out test dataset showed AUROC ranging from 0.81 (95% CI 0.73–0.89) to 0.87 (95% CI 0.80–0.93) and F1 scores ranging 0.74 (95% CI 0.64–0.83) to 0.78 (95% CI 0.69–0.86). ROC analysis for four models using the selected feature set demonstrated stable results for model training across all cross-validation folds (Supplementary Fig. 4). Figure 3 shows the comparative ROC analysis for testing the four classification models on the hold-out test dataset using the selected feature set and the model using SOFA scores alone. Extra Trees Classifier showed the best overall testing performance with AUROC: 0.87 (95% CI 0.80–0.93), accuracy: 0.79 (95% CI 0.71–0.86), F1 score: 0.78 (95% CI 0.69–0.86), precision: 0.88 (95% CI 0.78–0.98) and recall: 0.70 (95% CI 0.57–0.82). While using SOFA score alone, the model showed a comparatively lower testing AUROC of 0.77.

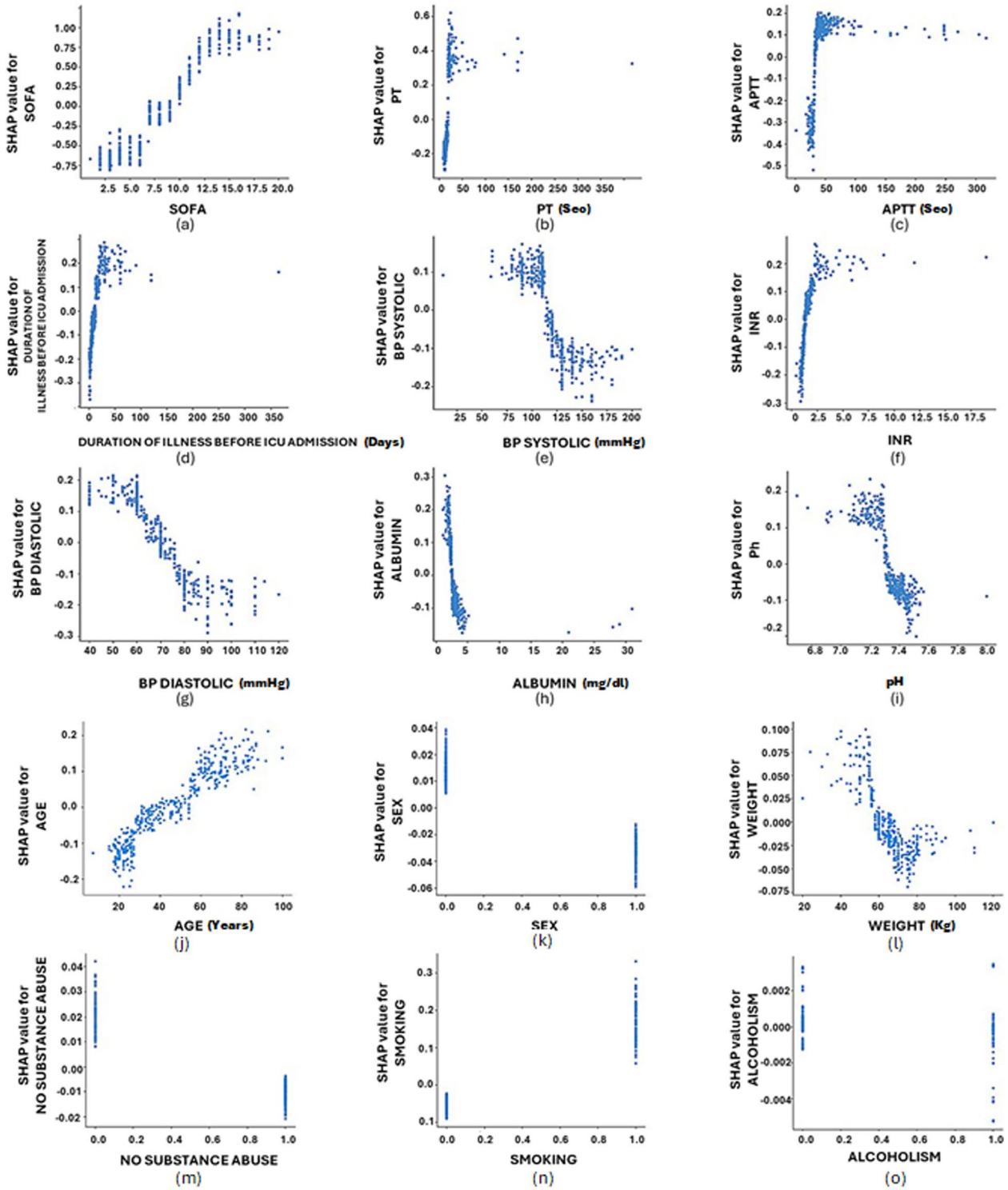


Fig. 2 SHAP-based dependence plots of most influential clinical features such as **a** SOFA scores, **b** prothrombin time (PT), **c** activated partial thromboplastin time (APTT), **d** duration of illness before ICU admission, **e** blood pressure (BP) systolic, **f** international normalized ratio (INR), g blood pressure (BP) diastolic, **h** albumin and **i** pH and demographic features such as **j** age, **k** sex, **l** weight, **m** no substance abuse, **n** smoking, and **o** alcoholism. Each point in the plot corresponds to an individual patient and the position of a said point along the x-axis corresponds to the value of the feature and along y-axis corresponds to the SHAP value for the feature under consideration. Please note that the scale of the y-axis is not the same for all plots; this was set to be able to capture the effects of all top features

Table 2 Comparative modeling analysis for selected features

ML models	Feature used	Fivefold cross-validation					Hold-out test				
		AUROC	Accuracy	F1 score	Precision	Recall	AUROC	Accuracy	F1 score	Precision	Recall
Random Forest Classifier	SHAP-based features	0.832 ± 0.019	0.758 ± 0.034	0.757 ± 0.039	0.744 ± 0.032	0.745 ± 0.054	0.86 (95% CI 0.79–0.93)	0.79 (95% CI 0.70–0.87)	0.78 (95% CI 0.68–0.86)	0.90 (95% CI 0.80–0.98)	0.69 (95% CI 0.55–0.81)
	SOFA score alone	0.786 ± 0.041	0.720 ± 0.042	0.723 ± 0.049	0.716 ± 0.067	0.734 ± 0.094	0.77 (95% CI 0.66–0.86)	0.68 (95% CI 0.59–0.77)	0.67 (95% CI 0.56–0.77)	0.75 (95% CI 0.62–0.87)	0.61 (95% CI 0.48–0.73)
XGBoost Classifier	SHAP-based features	0.843 ± 0.018	0.785 ± 0.038	0.779 ± 0.042	0.785 ± 0.043	0.775 ± 0.064	0.84 (95% CI 0.76–0.91)	0.75 (95% CI 0.66–0.83)	0.75 (95% CI 0.65–0.84)	0.82 (95% CI 0.70–0.93)	0.69 (95% CI 0.56–0.80)
	SOFA score alone	0.792 ± 0.043	0.720 ± 0.042	0.718 ± 0.040	0.716 ± 0.067	0.734 ± 0.094	0.77 (95% CI 0.68–0.85)	0.68 (95% CI 0.58–0.77)	0.67 (95% CI 0.56–0.77)	0.75 (95% CI 0.62–0.87)	0.61 (95% CI 0.48–0.74)
Extra Trees Classifier	SHAP-based features	0.841 ± 0.016	0.738 ± 0.042	0.726 ± 0.025	0.777 ± 0.035	0.709 ± 0.046	0.87 (95% CI 0.80–0.93)	0.79 (95% CI 0.71–0.86)	0.78 (95% CI 0.69–0.86)	0.88 (95% CI 0.78–0.98)	0.70 (95% CI 0.57–0.82)
	SOFA score alone	0.791 ± 0.043	0.727 ± 0.051	0.723 ± 0.049	0.727 ± 0.069	0.729 ± 0.085	0.77 (95% CI 0.67–0.86)	0.68 (95% CI 0.58–0.77)	0.67 (95% CI 0.56–0.77)	0.75 (95% CI 0.61–0.87)	0.61 (95% CI 0.48–0.74)
Gradient Boosting Classifier	SHAP-based features	0.822 ± 0.015	0.750 ± 0.021	0.750 ± 0.014	0.735 ± 0.035	0.735 ± 0.035	0.81 (95% CI 0.73–0.89)	0.76 (95% CI 0.67–0.85)	0.74 (95% CI 0.64–0.83)	0.88 (95% CI 0.76–0.97)	0.65 (95% CI 0.52–0.77)
	SOFA score alone	0.791 ± 0.044	0.720 ± 0.042	0.718 ± 0.040	0.716 ± 0.067	0.734 ± 0.094	0.77 (95% CI 0.67–0.85)	0.68 (95% CI 0.59–0.77)	0.67 (95% CI 0.56–0.77)	0.75 (95% CI 0.62–0.87)	0.61 (95% CI 0.49–0.74)

Table shows mean ± standard deviation of results of fivefold cross-validation for selected features and model performance on hold-out test set with 95% confidence Intervals

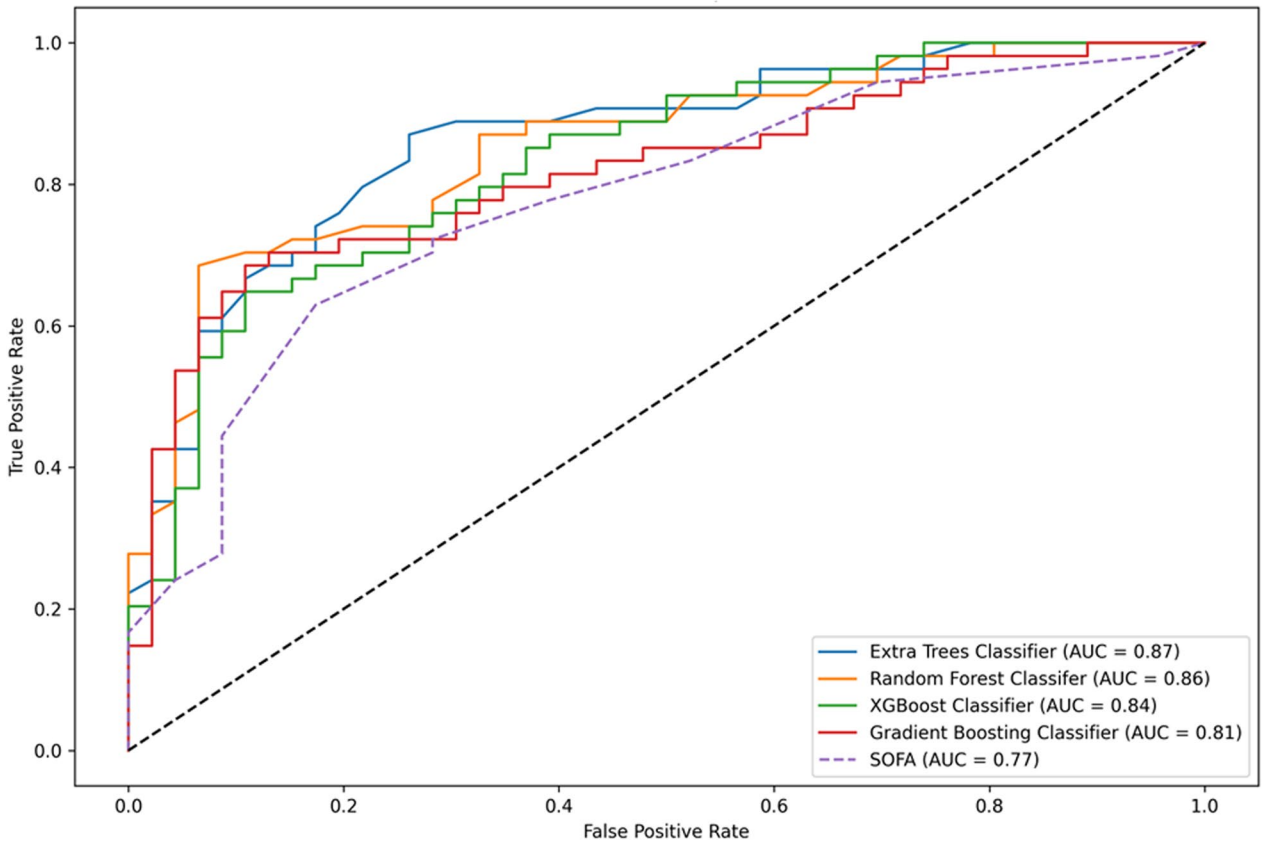


Fig. 3 Plots show AUROC curves for model testing on the hold-out test dataset to compare the performance using SHAP-based selected feature set with Extra Trees Classifier, Random Forest Classifier, XGBoost Classifier, and Gradient Boosting Classifier and model using only SOFA score as a feature

Discussion

The current study was performed to predict in-hospital mortality by using an explainable ML model in patients with sepsis admitted to the ICU. The study utilized demographic, clinical, radiological, and laboratory data of the patients admitted to the ICU for model development and validation. Even though the literature already has multiple machine learning models to predict mortality in sepsis [17], a relatively smaller number of models were developed in LMICs and LICs wherein as the demographic and clinical characteristics of patients are different from those in HICs. Therefore, the current study is of vital importance to capture the realistic and reproducible scenario in the subcontinent population. In this study all four ML models performed reasonably better (AUROC: 0.81–0.87) on the current dataset for predicting sepsis-related mortality than the current standard of practice using the severity scores SOFA alone (AUCOR: 0.77); while proposed Extra Trees Classifier exhibited the best overall performance with AUROC of 0.87.

Numerous studies in past have focused on prediction of in-hospital sepsis-related mortality showcasing a growing utility of different machine learning (ML) and deep learning (DL) algorithms, where RF and XGBoost were reported as the most preferable models [17]. In the developmental work for morality prediction models in sepsis among several ML algorithms XGBoost model has shown the best performance as reported by Bao et al. (AUROC:0.96) [21], Lu et al. (AUROC: 0.842) [22], and Choi et al. (AUROC: 0.977) [15]. On the other hand, among different ML models for mortality prediction, RF model also reported to produce the highest performance by Poucke et al. (AUROC: 0.74) [23], Kong et al. (AUROC:0.845) [24] in ICU patients and by Greco et al. [16], Taylor et al. [25] in emergency department with sepsis patients (AUROC: 0.86). Zhang et al. reported, LASSO score as the best discrimination with AUROC of 0.772 for in-hospital mortality in severe sepsis [26]. Choi et al. reported using MLP an AUROC of 0.835 for predicting mortality in sepsis [27]. A novel Quotient Basis Kernel for SVM was developed for mortality prediction in sepsis using soft-margin SVM and showed ~80% prediction accuracy [28]. Shin et al. leveraged word representations and embeddings out of clinical notes and along with

clinical data for early prediction of mortality and showed the best AUROC of 0.85 [29]. Additionally, Shukla et al. proposed a blockchain-based clinical decision support system for similar prediction and reported AUROC of 0.79 [30]. DL-based prediction models for sepsis-related mortality were developed by Yong et al. using deep neural network and graph convolutional network [31] and by Cheng et al. [32] using CNNs, LSTM and demonstrated an improved results producing than the ML-based prediction models. In comparison to the reported ML-based models for mortality prediction in sepsis, proposed ML models have a higher predictive accuracy than the models developed by [26–28, 30, 31]. Proposed RF model (AUROC: 0.86) demonstrated a better performance on the in-hospital dataset than reported by Poucke et al. (AUROC: 0.74) [23] and Kong et al. (AUROC:0.845) [24] in ICU patients; however, was comparable with RF models as reported by Greco et al. [16], Taylor et al. [25] and Cheng et al. [32] in Emergency department with sepsis patients (AUROC: 0.86). Proposed XGBoost model (AUROC: 0.84) showed a comparable accuracy with [22]; however, had less predictive accuracy than as reported by Bao et al. (AUROC: 0.96) [21] and Choi et al. (AUROC: 0.977) [15]. This may be due to variability in the feature set included in this study while Bao et al. used MIMIC-IV data containing only electronic medical records on patients in the United States [21] and Choi et al. used only limited clinical parameters [15]. In our study, Extra Trees Classifier demonstrated a higher accuracy than the widely used RF and XGBoost models because using the entire dataset during the model training Extra Trees allows to reduce the bias in the model compared to the RF and XGBoost models.

Overall ICU mortality in our study was 41.5% and discharge rate (Discharged to ward) was 58.5%. Mortality in this study population was lower than the ICU mortality of 56.1% reported by Chatterjee et al. [45], and slightly higher to the ICU mortality of 36.7% in a multicenter study of 150 ICUs from 16 Asian countries [46]. We divided the 500 patients into two groups based on binary outcomes (discharge to the ward and death in the ICU). No significant difference was observed between the two groups in the demographic parameters like, age and weight ($p = 0.3344$ and 0.5327). However, number of male patients ($p = 0.003$), patients with more substance abuse ($p = 0.003$), smoking ($p = 0.001$), and alcoholism ($p = 0.015$) were significantly higher in the mortality group than the discharge group similar to the previous studies [47–49]. In the current study, SOFA score, PT, and APTT were found to be the highest influential parameters followed by the duration of illness before ICU admission, systolic and diastolic blood pressure at the time of ICU admission to predict mortality among ICU patients. Similar observations were reported in [22] using SHAP values for feature importance. After SHAP analysis, while the SOFA score was observed to capture the highest contributing feature, however, its two sub-components, the absolute value of bilirubin and creatinine, were also found to be informative and contributed to the model performance. Though multicollinearity among SOFA and its sub-components may not be completely ruled out, yet it is not easy to determine because of the high non-linearity of the sub-scoring systems and the range of individual components contributing to determine a single consolidated and discrete SOFA score. Further, causal relations of the demographic and clinical parameters with risk of mortality among septicemia patients were observed similar as the earlier study by Zhang et al. [50]. A higher value of SOFA, PT, and APTT were found to be associated with a higher risk of mortality, while a lower systolic and diastolic blood pressure were associated with an increasing risk of mortality in patients with sepsis. Among demographic parameters, higher age and weight were observed to be associated with a high risk of mortality, while no substance abuse, no smoking, and non-alcoholism were associated with a lower risk of mortality and are also reported earlier [47–49]. 'AGE' was found to be an important feature after SHAP analysis, however at a lower rank compared to SOFA scores that hold a very high SHAP importance for predicting mortality due to sepsis. However, after removing SOFA scores from the feature list, 'AGE' showed high SHAP importance, although overall accuracy of the model reduced as SOFA was observed to be an essential feature for model functioning.

In the case of ICU patients, a wide array of data parameters is generated due to continuous monitoring. In such cases, making accurate decisions gets difficult for the clinician and, to an extent, for the machine learning algorithms, owing to their very high dimensionality. The proposed explainable feature selection methodology led us to reduce our dataset of 138 patient parameters to a key 39 parameters, representing 80% of the total spread of influence on the model output while retaining high model performance. The added data insights from such explainable predictive models in healthcare settings can help increase the trust of clinicians and drastically improve the adoption of AI-enhanced clinical decision support systems, putting them in a position of proactive care rather than reactive care delivery. This could in turn improve patient outcomes, quality of care, streamline treatment protocols, help allocate resources in an efficient manner, identify high-risk patients, develop new treatment modalities, and evaluate their efficacy [51]. This also paves the way for future modeling efforts by giving insight into what the model actually learns from

the data. With the help of such approaches, unexplored patterns not so visible to clinicians, as well as traditional statistical approaches, can be unearthed. In summary, the novel contributions of this study are twofold explainable AI-based investigation into Intensive Care Unit patients' data as well as data modeling using SHAP explanations. This encompassed identification of most influential features to the model's predictive outputs as well as looking into local and global inter-feature dependencies learnt by the model. Further, a robust machine learning-based classification model approach for high dimensional clinical parameters of mixed data type. Finally, development of predictive models to provide preemptive estimations into prognostication and outcome prediction of patients with sepsis in low-resource settings.

There are few limitations in this study. While this study displays a promising role for explainable AI-based approaches, especially in low-resource settings, generalizability is still a challenge. The dataset in the current study is limited as compared to those in the literature, such as the MIMIC dataset [33]. To facilitate adoption in clinical settings, large-scale multi-centric model validations as well as prospective studies are required while monitoring longitudinal effects. The proposed ML model is shared as an open-source tool through GitHub (https://github.com/amitvmehndiratta/eML_Sepsis_Prediction_ICME2025) and thus it can be accessible to larger community for independent evaluation. Other sources of clinical information in the form of text, image, waveform, etc., could not be included in the proposed prediction model due to limited resources and non-availability of EMR that can be assessed in retrospective studies, which might be a common case in the low-resource settings. However, more elaborated data can add value for generalizability of the model and thus must be included in future prospective studies. The current study works only with static data. Data points from time of admission to the ICU and interventions provided have been considered; however, time series data that have not been taken into account might have further improved the performance of the models [21]. The empirical insights observed via SHAP also require prospective data to note causality. A comparative analysis using SOFA score alone was performed for prediction mortality among septicemia patients, however another commonly used risk stratification score, APACHE II, could not be performed due to insufficient data and this may be taken up in the future work.

Conclusions

This study demonstrated the SOFA score, PT, APTT, systolic and diastolic blood pressure at the time of ICU admission and the duration of illness before ICU admission were the highest influence parameters for predicting death among patients with sepsis in ICU. With the explainable feature insights proposed ML models, specifically Extra Trees Classifier model showed satisfactory performance to stratify high-risk sepsis patients and can aid in clinical decision-making, treatment planning and resource management. A further study on prospective validation of the developed model is required, to evaluate predictive performance of the model.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s40635-025-00765-5.

Additional file 1

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Author contributions
V. G., P. K. and A. M. conceptualized the study. Data were curated by V. G. Methodology of the study was developed by S. D., E. B. K. and A. M. A. M. and P. K. supervised the study and allocated the resources. Formal analysis and investigations were performed by S. D. and E. B. K. Original manuscript and visualizations were prepared by S. D. and E. B. K. All authors reviewed and edited the manuscript.

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Availability of data and materials
The datasets generated and analyzed during the current study are not publicly available due institutional data confidentiality policy but are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
Study protocol was approved by the Institute Ethic Committee. Complete study protocol was explained, and written consent was obtained from all patients before participation.

Consent for publication
Not applicable

Competing interests
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

1. Pieroni M, Olier I, Ortega-Martorell S, Johnston BW, Welters ID (2022) In-hospital mortality of sepsis differs depending on the origin of infection: an investigation of predisposing factors. Front Med. https://doi.org/10.3389/fmed.2022.915224
2. Rhee C, Dantes R, Epstein L, Murphy DJ, Seymour CW, Iwashyna TJ, Kadri SS, Angus DC, Danner RL, Fiore AE, Jernigan JA, Martin GS, Septimus E, Warren DK, Karcz A, Chan C, Menchaca JT, Wang R, Gruber S, Klompas M (2017) Incidence and trends of sepsis in US hospitals using clinical vs claims data, 2009–2014. JAMA 318:1241–1249. https://doi.org/10.1001/JAMA.2017.13836
3. Rudd KE, Johnson SC, Agesa KM, Shackelford KA, Tsoi D, Kievlan DR, Colombara DV, Ikuta KS, Kissoon N, Finfer S, Fleischmann-Struzek C, Machado FR, Reinhart KK, Rowan K, Seymour CW, Watson RS, West TE, Marinho F, Hay SI, Lozano R, Lopez AD, Angus DC, Murray CJL, Naghavi M (2020) Global, regional, and national sepsis incidence and mortality, 1990–2017: analysis for the global burden of disease study. Lancet 395:200–211. https://doi.org/10.1016/S0140-6736(19)32989-7
4. Gaieski DF, Edwards JM, Kallan MJ, Carr BG (2013) Benchmarking the incidence and mortality of severe sepsis in the United States. Crit Care Med 41:1167–1174. https://doi.org/10.1097/CCM.0B013E31827C09F8
5. Klompas M, Rhee C (2016) Sepsis and the theory of relativity: measuring a moving target with a moving measuring stick. Crit Care 20:1–3. https://doi.org/10.1186/S13054-016-1559-Z/METRICS
6. Shankar-Hari M, Phillips GS, Levy ML, Seymour CW, Liu VX, Deutschman CS, Angus DC, Rubenfeld GD, Singer M (2016) Developing a new definition and assessing new clinical criteria for septic shock: for the third international consensus definitions for sepsis and septic shock (Sepsis-3). JAMA 315:775–787. https://doi.org/10.1001/JAMA.2016.0289
7. Rhee C, Gohil S, Klompas M (2014) Regulatory mandates for sepsis care—reasons for caution. N Engl J Med 370:1673–1676. https://doi.org/10.1056/NEJMP1400276
8. Jafarzadeh SR, Thomas BS, Marschall J, Fraser VJ, Gill J, Warren DK (2016) Quantifying the improvement in sepsis diagnosis, documentation, and coding: the marginal causal effect of year of hospitalization on sepsis diagnosis. Ann Epidemiol 26:66–70. https://doi.org/10.1016/J.ANNEPIDEM.2015.10.008
9. Li A, Ling L, Qin H, Arabi YM, Myatra SN, Egi M, Kim JH, Basri M, Nor M, Son N, Fang W-F, Wahyuprajitno B, Hashmi M, Faruq MO, Patjanasoontorn B, Jaffer M, Bahrani A, Shrestha BR, Shrestha U, Mahmood K, Nafees K, Sann KK, Emmanuel J, Palo M, Mendsaikhan N, Konkayev A, Detleuxay K, Chan YH, Du B, Divatia JV, Koh Y, Gomersall CD, Phua J (2022) Epidemiology, management, and outcomes of sepsis in ICUs among countries of differing national wealth across Asia. Am J Respir Crit Care Med. https://doi.org/10.1164/rccm.202112-2743OC
10. Jeganathan N (2022) Burden of sepsis in India. Chest 161:1438–1439. https://doi.org/10.1016/j.chest.2022.02.008
11. Vincent J-L, Moreno R, Takala J, Willatts S, de Mendona A, Bruining H, Reinhart CK, Suter PM, Thijs LG, De Mendonga A, Suter RM (1996) The SOFA (Sepsis.related organ failure assessment) score to describe organ dysfunction/failure. Intensive Care Med 22:707–710
12. Gil Cebrian J, Bello Cámara MP, Diaz-Alersi R (1987) APACHE II: a severity of disease classification system. Intensive Care Med 13:143. https://doi.org/10.1097/00003465-198603000-00013
13. Do SN, Dao CX, Nguyen TA, Nguyen MH, Pham DT, Nguyen NT, Huynh DQ, Hoang QTA, Van Bui C, Vu TD, Bui HN, Nguyen HT, Hoang HB, Le TTP, Nguyen LTB, Duong PT, Nguyen TD, Le VH, Pham GTT, Bui GTH, Van Bui T, Pham TTN, Van Nguyen C, Nguyen AD, Phua J, Li A, Luong CQ (2023) Sequential organ failure assessment (SOFA) score for predicting mortality in patients with sepsis in Vietnamese intensive care units: a multicentre, cross-sectional study. BMJ Open. https://doi.org/10.1136/bmjopen-2022-064870
14. Mirijello A, Fontana A, Greco AP, Tosoni A, D'Agruma A, Labonia M, Copetti M, Piscitelli P, De Cosmo S (2023) on behalf of the internal medicine sepsis study group, identifying predictors associated with risk of death or admission to intensive care unit in internal medicine patients with sepsis: a comparison of statistical models and machine learning algorithms. Antibiotics. https://doi.org/10.3390/antibiotics12050925
15. Choi MH, Kim D, Choi EJ, Jung YJ, Choi YJ, Cho JH, Jeong SH (2022) Mortality prediction of patients in intensive care units using machine learning algorithms based on electronic health records. Sci Rep. https://doi.org/10.1038/s41598-022-11226-4
16. Greco M, Caruso PF, Spano S, Citterio G, Desai A, Molteni A, Aceto R, Costantini E, Voza A, Cecconi M (2023) Machine learning for early outcome prediction in septic patients in the emergency department. Algorithms. https://doi.org/10.3390/a16020076
17. Zhang Y, Xu W, Yang P, Zhang A (2023) Machine learning for the prediction of sepsis-related death: a systematic review and meta-analysis. BMC Med Inform Decis Mak. https://doi.org/10.1186/s12911-023-02383-1
18. Petch J, Di S, Nelson W (2022) Opening the black box: the promise and limitations of explainable machine learning in cardiology. Can J Cardiol 38:204–213. https://doi.org/10.1016/J.CJCA.2021.09.004
19. Amann J, Blasimme A, Vayena E, Frey D, Madai VI (2020) Explainability for artificial intelligence in healthcare: a multidisciplinary perspective. BMC Med Inform Decis Mak 20:310. https://doi.org/10.1186/s12911-020-01332-6
20. Reddy S (2022) Explainability and artificial intelligence in medicine, lancet digit. Health 4:e214–e215. https://doi.org/10.1016/S2589-7500(22)00029-2
21. Bao C, Deng F, Zhao S (2023) Machine-learning models for prediction of sepsis patients mortality. Medicina Intensiva 47:315–325. https://doi.org/10.1016/j.medine.2022.06.024
22. X. Lu, J. Zhu, J. Gui, Q. Li, Prediction of All-cause Mortality with Sepsis-associated Encephalopathy in the ICU Based on Interpretable Machine Learning, in: 2022 IEEE International Conference on Mechatronics and Automation, ICMA 2022, Institute of Electrical and Electronics Engineers Inc., 2022: pp. 298–302. https://doi.org/10.1109/ICMA54519.2022.9856126.
23. van Poucke S, Zhang Z, Schmitz M, Vukicevic M, Vander Laenen M, Celi LA, de Deyne C (2016) Scalable predictive analysis in critically ill patients using a visual open data analysis platform. PLoS One. https://doi.org/10.1371/journal.pone.0145791
24. Kong G, Lin K, Hu Y (2020) Using machine learning methods to predict in-hospital mortality of sepsis patients in the ICU. BMC Med Inform Decis Mak. https://doi.org/10.1186/s12911-020-01271-2
25. Taylor RA, Pare JR, Venkatesh AK, Mowafi H, Melnick ER, Fleischman W, Hall MK (2016) Prediction of in-hospital mortality in emergency department patients with sepsis: a local big data-driven, machine learning approach. Acad Emerg Med 23:269–278. https://doi.org/10.1111/acem.12876
26. Z. Zhang, Y. Hong, Development of a novel score for the prediction of hospital mortality in patients with severe sepsis: the use of electronic healthcare records with LASSO regression, 2017. www.impactjournals.com/oncotarget.
27. J.W. Choi, J.W. Kim, J.H. Nam, J.Y. Maeng, K.H. Kim, S. Park, Artificial Intelligence for Predicting Mortality Due to Sepsis, in: Digest of Technical Papers - IEEE International Conference on Consumer Electronics, Institute of Electrical and Electronics Engineers Inc., 2023. https://doi.org/10.1109/ICCE56470.2023.10043540.
28. Ribas Ripoll VJ, Vellido A, Romero E, Ruiz-Rodríguez JC (2014) Sepsis mortality prediction with the quotient basis kernel. Artif Intell Med 61:45–52. https://doi.org/10.1016/j.artmed.2014.03.004
29. J. Shin, Y. Li, Y. Luo, Early Prediction of Mortality in Critical Care Setting in Sepsis Patients Using Structured Features and Unstructured Clinical Notes, Proceedings - 2021 IEEE International Conference on Bioinformatics and Biomedicine, BIBM 2021 (2021) 2885–2890. https://doi.org/10.1109/BIBM52615.2021.9669822.
30. M. Shukla, J. Lin, O. Seneviratne, Collaboratively Learning Optimal Patient Outcomes Using Smart Contracts in Limited Data Settings, in: Proceedings - 2022 IEEE/ACM International Conference on Connected Health: Applications, Systems and Engineering Technologies, CHASE 2022, Institute of Electrical and Electronics Engineers Inc., 2022: pp. 133–137. https://doi.org/10.1145/3551455.3559600.
31. Yong L, Zhenzhou L (2024) Deep learning-based prediction of in-hospital mortality for sepsis. Sci Rep. https://doi.org/10.1038/s41598-023-49890-9
32. Cheng C-Y, Kung C-T, Chen F-C, Chiu I-M, Lin C-HR, Chu C-C, Kung CF, Su C-M (2022) Machine learning models for predicting in-hospital mortality in patient with sepsis: analysis of vital sign dynamics. Front Med. https://doi.org/10.3389/fmed.2022.964667
33. Johnson AEW, Pollard TJ, Shen L, Lehman LWH, Feng M, Ghassemi M, Moody B, Szolovits P, Anthony Celi L, Mark RG (2016) MIMIC-III, a freely

accessible critical care database. Sci Data. https://doi.org/10.1038/sdata.2016.35
34. Steinmeyer C, Wiese L (2020) Sampling methods and feature selection for mortality prediction with neural networks. J Biomed Inform 111:103580. https://doi.org/10.1016/J.JBI.2020.103580
35. Mazurowski MA, Habas PA, Zurada JM, Lo JY, Baker JA, Tourassi GD (2008) Training neural network classifiers for medical decision making: the effects of imbalanced datasets on classification performance. Neural Netw 21:427–436. https://doi.org/10.1016/j.neunet.2007.12.031
36. Aushev A, Ripoll VR, Vellido A, Aletti F, Pinto BB, Herpain A, Post EH, Medina ER, Ferrer R, Baselli G, Bendjelid K (2018) Feature selection for the accurate prediction of septic and cardiogenic shock ICU mortality in the acute phase. PLoS ONE 13:e0199089. https://doi.org/10.1371/JOURNAL.PONE.0199089
37. Remeseiro B, Bolon-Canedo V (2019) A review of feature selection methods in medical applications. Comput Biol Med 112:103375. https://doi.org/10.1016/j.compbiomed.2019.103375
38. S.M. Lundberg, P.G. Allen, S.-I. Lee, A Unified Approach to Interpreting Model Predictions, Adv Neural Inf Process Syst 30 (2017). https://github.com/slundberg/shap (accessed April 17, 2024).
39. Shankar-Hari M, Phillips GS, Levy ML, Seymour CW, Liu VX, Deutschman CS, Angus DC, Rubenfeld GD, Singer M (2016) Developing a new definition and assessing new clinical criteria for septic shock: for the third international consensus definitions for sepsis and septic shock (sepsis-3). JAMA - J Am Med Assoc 315:775–787. https://doi.org/10.1001/jama.2016.0289
40. J. Krause, A. Perer, K. Ng, Interacting with Predictions: Visual Inspection of Black-box Machine Learning Models, (n.d.). https://doi.org/10.1145/2858036.2858529.
41. Nielsen RL, Monfeuga T, Kitchen RR et al (2024) Data-driven identification of predictive risk biomarkers for subgroups of osteoarthritis using interpretable machine learning. Nat Commun 15:2817. https://doi.org/10.1038/s41467-024-46663-4
42. Lundberg SM, Erion G, Chen H, DeGrave A, Prutkin JM, Nair B, Katz R, Himmelfarb J, Bansal N, Lee SI (2020) From local explanations to global understanding with explainable AI for trees. Nat Mach Intell 2:56–67. https://doi.org/10.1038/s42256-019-0138-9
43. T. Chen, C. Guestrin, XGBoost: A Scalable Tree Boosting System, in: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, Association for Computing Machinery, New York, NY, USA, 2016: pp. 785–794. https://doi.org/10.1145/2939672.2939785.
44. Shwartz-Ziv R, Armon A (2022) Tabular data: deep learning is not all you need. Information Fusion 81:84–90. https://doi.org/10.1016/J.INFFUS.2021.11.011
45. Chatterjee S, Bhattacharya M (2017) Epidemiology of adult-population sepsis in India: a single center 5 year experience, Indian. J Crit Care Med 21:573–577. https://doi.org/10.4103/ijccm.ijccm_240_17
46. Divatia JV, Amin PR, Ramakrishnan N, Kapadia FN, Todi S, Sahu S, Govil D, Chawla R, Kulkarni AP, Samavedam S, Jani CK, Rungta N, Samaddar DP, Mehta S, Venkataraman R, Hegde A, Bande BD, Dhanuka S, Singh V, Tewari R, Zirpe K, Sathe P, Sivakumar MN, Kansal S, Gurav S, Kulkarni S, Ray S, Rajhans P, Choudhary S, Singhvi T, Toraskar K, Pande R, Gupta S, Rao SM, Kujore R, Kumar Mani A, Sinha V, Kulkarni A, Moulick ND, Shah C, Lakhani J, Myatra SN, Sircar M, Shah RJ, Singh DK, Shenoy A, Dubey PK, Joshi K, Dewan A, Kumar V, Goel C, Tyagi RS, Bhutada K, Mishra S, Pasha SA, Nangia V, Deka AK, Munjal M, Arunkumar AS, Chavda MP, Hartalkar AR, Kar A, Shrivas VK, Sarna MK, Singh AK, Sahasrabudhe S, Daniel B, Sharma S, Agrawal A, Gopinath R, Sathiakumar DBD, Kadapatti K, Sharma J, Lahkar B, Chaudhry D, Azim A, Prayag S, Talekar S, Patil S, Tipparaju SS, Kola VR, Ashok E, Joad SHK, Jeswani D, Bose N, Mannari JG, Karnad DR, Clerk A, Pattnaik S, Shanbhag V, Barokar R, Nakra CM, Venugopal M, Ramasubban S, Gupta A, Gupta B, Aditya T, Prasad R, Joshi MM, Dey A, Kulkarni V, Sunder S, Bhavsar A, David E, Jaju M, Pandya ST, Balasubramani VM, Dinaker M, Khunteta S, Shah H, Srinivasulu B, Gautam PL, Sood P, Acharya K, Jindal VB, Yadgire R, Bhattacharyya S, John JK, Nanda A, Giri S, Oza PL, Shah J, Aghara D, Singh M, Sahu SP, Patil P, Tulara N, Chafle S, Timane J, Tandon K, Benjarge PV, Thakur VK, Mehta J, Singhvi GR, Parashar VK, Malani A, Sivakumar MN, Singh DK, Singh AK, Gopinath R, Sathiakumar DBD, Ashok E, Venugopal M, Aditya T, Balasubramani VM, Dinaker M, Srinivasulu B, Gautam PL, Jindal VB, Thakur VK, Singhvi GR, Parashar VK (2016) intensive care in India: the Indian intensive care case mix and practice patterns study. Ind J Crit Care Med 20:216–225. https://doi.org/10.4103/0972-5229.180042
47. Jabri A, Fowler C, Dhamija Y, Alzubi J, Bhatia S, Al-abdouh A, Alameh A, Alfahel H, Haddadin F, Shahrori Z, Nasser F, Ababneh A (2022) Demographic undertones for sepsis mortality in a community-based hospital. J Clin Med Res 14:28–33. https://doi.org/10.1474/jocmr4618
48. Stenberg H, Li X, Pello-Esso W, Larsson Lönn S, Thønnings S, Khoshnood A, Knudsen JD, Sundquist K, Jansåker F (2023) The effects of sociodemographic factors and comorbidities on sepsis: a nationwide Swedish cohort study. Prev Med Rep. https://doi.org/10.1016/j.pmedr.2023.102326
49. Lindström AC, Eriksson M, Mårtensson J, Oldner A, Larsson E (2021) Nationwide case–control study of risk factors and outcomes for community-acquired sepsis. Sci Rep. https://doi.org/10.1038/s41598-021-94558-x
50. Zhang Z, Jin P, Feng M et al (2022) Causal inference with marginal structural modeling for longitudinal data in laparoscopic surgery: a technical note. Laparosc Endosc Robot Surg 5(4):146–152. https://doi.org/10.1016/j.lers.2022.10.002
51. Boussina A, Shashikumar SP, Malhotra A, Owens RL, El-Kareh R, Longhurst CA, Quintero K, Donahue A, Chan TC, Nemati S, Wardi G (2024) Impact of a deep learning sepsis prediction model on quality of care and survival. NPJ Digit Med 7:14. https://doi.org/10.1038/s41746-023-00986-6

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